



Management of complications after appendectomy: Literature review



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Introduction

Acute appendicitis is one of the most prevalent surgical entities at emergency department, and has an overall lifetime risk of 7%-8%.¹ Although it can be discussed whether to operate or not on these patients, its current mainstay therapy is appendectomy and it is the most commonly performed procedure within an acute care surgery service.¹ While the mortality rate after appendectomy has decreased significantly in recent years, the morbidity rate still remains substantial.

Patients with acute appendicitis are classified as uncomplicated or complicated based on pre-operative, intra-operative, and/or histopathological findings. Uncomplicated appendicitis is defined as an inflamed appendix without signs of gangrene, perforation or intra-peritoneal purulent fluid. As a counterpart, complicated appendicitis applies to all cases with either a gangrenous inflamed appendix with or without perforation, intraabdominal abscess, peri-appendicular contained phlegmon, or purulent free fluid. Classification is necessary because treatment strategies might differ.² Both open and laparoscopic approach are commonly used for appendectomy, and the implementation of one over the other depends on multiple variables such as patient's condition, type of appendicitis and surgical team expertise among others. Several studies³⁻⁶ have shown the benefits of laparoscopic appendectomy as compared to the open approach, such as shorter hospital stay, less postoperative pain, earlier return to daily activities and lower wound infection rate. Furthermore, patients showed a strong preference for minimally invasive surgery and high satisfaction after surgery. Bondi et al.⁴ suggests that with the required expertise and equipment laparoscopy should be considered the initial choice for most cases of suspected appendicitis. Independently of the approach, it is important to note that appendectomy is an urgent procedure that could develop several postoperative complications, therefore, the primary approach may determine how to manage each unfortunate event.

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Complications could be defined as any deviation from the postoperative during the first 30 postoperative days. Long-term complications as incisional hernias might also occur. Complications after appendectomy can occur in up to 17% of patients, including surgical site infection (SSI), intra-abdominal abscess (IAA), incisional hernias, ileus, bowel obstruction, bleeding, or stump appendicitis.⁷ The presence of any of them are often related to longer postoperative recovery and increased healthcare expenses.

In this review, we describe early and late complications that may arise after an appendectomy. Current strategies for their management are also discussed. We aimed to provide a comprehensive overview that could serve as a valuable resource for the care of patients undergoing appendectomy.

Type of complication and resolution

Intraabdominal abscess (IAA)

Intraabdominal abscess is the most feared complication after appendectomy, and its incidence is highly variable ranging between 4.2%–28%.^{8,9} IAA occurs more frequently among patients with complicated appendicitis and increases morbidity rates with longer hospital stay and needs of complementary treatment. In the presence of postoperative fever, abdominal pain and/or leukocytosis, an IAA should be suspected. In these patients an abdominal imaging such as ultrasound or CT scan should be performed to confirm the diagnosis. CT scan is the preferred modality since it provides the needed information for therapeutic guidance. Several studies have analyzed risk factors.^{8,10,11} for the development of IAA. Obesity, CRP levels greater than 200 mg/L, leukocyte count greater than 17,000/mm³, pelvic peritonitis, and perforated appendicitis are considered independent risk factors for postoperative IAA.

Upon the diagnosis of IAA, antibiotic therapy, percutaneous drainage, reoperation or a combination of them might be needed.^{12–14} Therapy can be decided based on the size, localization, and number of collections. A step-up management of IAA is mandatory to avoid unnecessary procedures.¹³ Conservative treatment with antibiotic therapy seems a reasonable option for one IAA smaller than 3 cm. On the other hand, isolated antibiotic therapy is much less effective when abscesses are larger than 3 cm in size. In those cases, percutaneous drainage by interventional radiology has become the treatment of choice. In general, percutaneous abscess drainage is performed in patients with fluid collections of more than 3 cm in diameter and appropriate clinical indications, with use of local anesthesia and sedation.¹⁵ Limitations of percutaneous drainage include cost, availability in hospitals and abscesses in locations that are difficult to access.¹⁶ Also, it is no free from risks such as bowel perforation, fistula formation, hemorrhage, and postprocedural sepsis.¹⁷

Cultures play an essential role in surgical and post-procedural management, particularly in cases involving IAA. Although there is no consensus regarding intraoperative cultures sampling, this step could be important for identifying the microorganisms responsible for abscess formation. This information guides the selection of appropriate antimicrobial therapy, ensuring targeted treatment and minimizing the risk of antibiotic resistance.¹⁸

Some patients with postoperative IAA might require surgical resolution such as those with multiple collections and/or percutaneous drainage failure.^{14,19} Although Casas et al. found that complications after appendectomy requiring a surgical therapy are infrequent, laparoscopy is an effective tool for diagnosis and treatment.¹⁹ Even though laparoscopic approach has several benefits in postoperative outcomes, some surgeons advocate open laparotomy with thorough irrigation when percutaneous drainage is ineffective or contraindicated. However, open approach has shown numerous drawbacks, including incisional pain, elevated rates of wound infections, risk of incisional hernias, and suboptimal cosmetic outcomes.²⁰ Laparoscopy offers the ability to explore the entire abdominal cavity and when necessary, drain multiple sites of abscesses.¹⁴ Additionally, some studies have demonstrated that laparoscopy results in reduced pain, shorter length

of stay, faster return to normal activity, decreased postoperative adhesions, and lower wound infection rates.²¹

Overall, a step-up approach seems to be the most reasonable strategy for the management of IAA. Beginning with no invasive strategies as the first approach, failure in treatment can go up to 16% for antibiotic therapy only, 12.5% for percutaneous drainage and 12.5% for surgery.¹³ Non-operative alternatives could be effective in most of the cases, leading to reoperation, preferable laparoscopic, in cases where percutaneous drainage is not feasible or has failed.

Surgical site infection

Surgical site infection (SSI) is one of the most common complications after appendectomy and its incidence reaches up to 11%.²² It can be a superficial infection involving the skin only or more serious infections, usually after the fifth postoperative day.²³ Laparoscopy has gained popularity in terms of preventing this complication, but it is still elevated. A multicenter prospective study has shown a reduction in wound infection rate when comparing open and laparoscopic approaches (8.7% vs 4.2%, respectively).²⁴ Nevertheless, a recent meta-analysis in pediatric patients did not find statistically significant differences in postoperative wound infections between laparoscopic and open appendectomy (odds ratio [OR], 0.63; 95% confidence interval [CI], 0.34–1.15; $P=0.13$).²⁵

Considering that appendectomy is not a clean procedure, the management of SSI begins with meticulous postoperative wound care, facilitating the prompt identification of any potential issues. Sterile dressing changes and constant monitoring for signs of infection, such as redness, leakage, and purulent discharge are vital. If cultures were obtained during the operation, is mandatory to adjust the antibiotic therapy as SSI is probably associated with the same microorganisms.¹⁸ For superficial wounds, drainage and antibiotics are the initial step in managing wound infections.²⁶ Otherwise, if no improvement is observed, a surgical debridement of the wound should be performed and the infectious department should also be consulted.

Appendix retrieval is probably a key step in preventing SSI since the passage of the specimen through the skin or ports could be a contamination source. In concordance, a study has demonstrated that the use of wound irrigation using wet gauze after fascia closure has a beneficial effect on reducing post-operative SSI.²⁷ On the other hand, Turner et al.,²⁸ performed a comparative analysis of the usefulness of retrieving bag in laparoscopic appendectomy. The authors included a total of 10,357 patients, and they found that the use of retrieval bag during laparoscopic appendectomy was not associated with a decrease in SSI rates.

SSI can generally be controlled with local wound management and antibiotics in most of cases. The relevance of this complication is not only due to its relatively high incidence, but also due to the negative impact in patient's recovery and return to daily activities. Therefore, despite being considered a minor complication, all alternatives in preventing SSI based on surgical team experience and common sense should be encouraged.

Ileus and bowel obstruction

Postoperative ileus or bowel obstruction after appendectomy is not a frequent complication. Nevertheless, they are reported as the third most common cause of unplanned readmission, with rates reaching up to 10%.²⁹ Employing minimally invasive techniques and avoiding unnecessary dissection maneuvers are essential to prevent these complications. In concordance, a comparative study analyzed the readmissions rate for small bowel obstruction after appendectomy comparing open and laparoscopic approaches. The authors found that the laparoscopic approach has significant advantages over the open technique (0.4% vs 1.0%, $P=0.0015$).³⁰

Imaging studies, preferably CT scan, can help distinguishing ileus from bowel obstruction. Additionally, CT can also rule out potential causes of ileus in which a specific treatment is required, such as intra-abdominal free fluid or IAA. Early mobilization, reduction in opioid use, and

the implementation of enhanced recovery after surgery (ERAS) protocols have shown promising results in preventing and managing ileus.³¹ Additionally, prokinetic agents can also accelerate the return of normal bowel function. Once the ileus is established, several treatments including bowel rest, fluid and ionic reposition, nasogastric decompression, chewing gum, drinking coffee or even re-laparoscopy have been described to address this complication. Ngowe et al.,³² found that chewing gum after open appendectomy stimulates bowel function, promoting faster oral intake and reducing length of hospital stay.

In cases of recurrent or persistent obstruction, surgical intervention might be necessary. Laparoscopy is emerging as the preferred approach in centers with extensive experience in minimally invasive techniques.³³ The use of laparoscopy for small bowel obstruction is controversial due to concerns about reduced working space and the potential for iatrogenic injuries resulting from bowel distention. However, laparoscopy should be considered in surgeons' armamentarium following the case-by-case basis.³⁴ In hemodynamically unstable patients with an acute abdomen, laparoscopy may not be the initial choice; on the contrary, it is strongly recommended for stable patients with non-resolving symptoms. Laparoscopic exploration and adhesiolysis, a key component for addressing adhesive small bowel obstruction, have been shown to be a safe alternative when conservative and non-operative measures fail. To optimize the success of laparoscopic adhesiolysis, several predictive factors have been recognized: history of ≤ 2 previous laparotomies, non-median incisions in previous laparotomies, prior appendectomy as the primary surgery leading to adhesions, the presence of a single band adhesion, prompt laparoscopic management within 24 hours of symptom onset, absence of peritonitis signs on physical examination, and surgeon's experience in performing laparoscopic procedures.^{35,36}

Overall, both ileus and bowel obstruction respond to several causes and their management requires an appropriate recognition and experience of the surgical team. Imaging, mainly CT scan, plays an important role in ruling out causes that need surgical intervention and it should not be delayed. In stable patients, laparoscopy allows diagnostic and treatment ensuring optimal outcomes for patients with adhesive small bowel obstruction.

Bleeding

Bleeding after appendectomy is rare but is a potentially serious complication that requires prompt and appropriate management. Potential areas of bleeding include the abdominal cavity, retroperitoneum, digestive tract, or port sites.³⁷ Gastrointestinal hemorrhage, occurring either early or years post-appendectomy, is typically managed through ligation of the bleeding vessel or cecal resection via emergency laparotomy or laparoscopy. In selected cases, angiographic embolization stands as an alternative consideration for effective management. The significance of early recognition and appropriate intervention cannot be overstated in mitigating the risks associated with post-appendectomy bleeding.

Few studies have described the incidence and treatment of post appendectomy bleeding. Among the literature, the reported incidence is 2.8%, although it is likely underreported.³⁸ The literature contains limited cases detailing patients experiencing postoperative bleeding after appendectomy.^{37,39,40} In cases where the bleeding is intraluminal, resolution can be achieved endoscopically. Instances of bleeding from an ulcer in the appendiceal stump have been documented in medical literature. This approach allows for direct visualization and targeted intervention, such as the application of hemoclips and the use of a biological tissue adhesive, providing a less invasive means to address the specific source of bleeding in the appendiceal stump.⁴⁰ On the other hand, when hemoperitoneum or retroperitoneal bleeding are evident, the appropriate course of action involves surgical intervention either through laparotomy or laparoscopy to address the source of bleeding and control. Thus, surgical exploration is crucial for managing cases where non-operative measures may not suffice to address the severity of the bleeding and its potential impact on the patient's overall condition.⁴¹

Finally, post-appendectomy bleeding is an unusual condition that is in most cases self-limited showing scarce GI hemorrhage or hematoma localized in the surgical area. However, it could

be life threatening and both endoscopic or surgical approaches must be quickly considered in patients who do not respond to medical resuscitation.

Incisional hernias

As in any other abdominal surgery, the risk of developing an incisional hernia (IH) after appendectomy is latent. It has been observed that up to 15% of patients could develop an IH after laparotomy and several risk factors have been described such as emergency surgery, hemorrhage, trauma, obesity, abdominal sepsis and type, and length of the incision among others.^{42,43} It is assumed that oblique incisions habitually used for open treatment of acute appendicitis has reduced risk for incisional hernia. In concordance, several studies have shown a rate lower than 2% for IH when a McBurney incision was implemented.^{44,45} In addition, Drs Beltran and Cruces⁴² performed a retrospective analysis in 4593 patients who undergo open appendectomy with McBurney incision and found that SSI, diabetes and female gender were independent risk factors for developing an incisional hernia.

The laparoscopic approach has shown to reduce IH rates in abdominal surgery since small incisions are used.^{46,47} However, against any thoughts, trocar site hernias (TSH) seem to be more frequent than expected ranging from 0.5%-2% depending on the type of surgery analyzed.⁴⁸ One potential explanation could be that the umbilical area is the most frequent localization of trocar placement and the umbilicus is a well-known risk factor for ventral IH.^{48,49} Furthermore, a misdiagnosed umbilical hernia and also the presence of rectus diastasis at the umbilical area could alter the wound healing process leading to IH development.^{48,50} A recent study conducted by Hernandez et al.⁵¹ showed an incidence of umbilical TSH after laparoscopic appendectomy of 20.3%, all of which were diagnosed with CT scan. This finding is alarming and highlights the necessity of IH prevention. For instance, the occurrence of TSH is reduced when employing non-bladed trocars. In addition, a key preventive measure involves the meticulous closure of all trocar sites larger than 5 mm. This meticulous closure serves as a proactive strategy to mitigate the risk of trocar site hernia formation.^{45,52,53}

Surgical repair is often considered the definitive treatment for incisional hernias. While McBurney incisional hernia is classically approached through open repair, it has been demonstrated that a minimally invasive surgery (MIS) utilizing a three-port technique and the placement of a composite mesh is equally effective for resolving the condition.⁵⁴ Similar outcomes have been observed for the treatment of ventral incisional hernia repairs in the midline via minimally invasive approach.⁵⁵ Nowadays, MIS in hernia repair has gained popularity and most hernia surgeons prefer minimally invasive techniques to manage mid-sized ventral hernias.^{55,56}

There is no doubt that IH and TSH after appendectomy are relevant complications, and major efforts are needed to prevent them. Once the hernia is established, a ventral hernia repair with mesh reinforcement with MIS is the suggested treatment approach.

Stump appendicitis

Stump appendicitis is a long-term complication after appendectomy that consists of the inflammation of the remaining appendix. The lifelong probability of developing stump appendicitis is low (1/50,000), representing a rare condition.⁵⁷ Signs and symptoms are the same as acute appendicitis and its diagnosis is usually delayed. Along the literature, the time between the incomplete appendectomy and stump appendicitis vary between 2 months and 20 years.⁵⁸ Risk factors for developing stump appendicitis are the presence of a complicated appendicitis at the initial appendectomy, improper identification of the appendicular base, and leaving a stump larger than 0.5 cm.^{19,59} After the suspicion arises, a CT scan is the method of choice for the correct diagnosis.

Surgery is often needed for treating this complication. Laparoscopic or open completion of the appendectomy is the adequate treatment.⁶⁰ Generally, a more extensive procedure should

not be required if the appendiceal stump is easily identified, and the cecum does not exhibit significant inflammation. Most reported cases in the literature involve open completion appendectomy, with ongoing debates about the inversion of the remaining stump versus simple ligation. Other authors suggest that, unless there are mitigating reasons, laparoscopic completion appendectomy guided by CT findings is a feasible approach.⁶¹ Manatakis et al.⁶¹ observed that delayed diagnosis of stump appendicitis contributed to the need for more extensive resections, including ileocectomy or right hemicolectomy in up to 11% of cases. The need for major resections is related to gangrene, perforation, or significantly inflammatory process of the cecum.

The use of minimally invasive techniques remains technically feasible, even following an initial open procedure in the hands of experienced surgeons. Laparoscopy allows for effective intervention while minimizing the invasiveness of the surgical procedure.

Conclusion

Appendectomy is a commonly performed surgical procedure that carries inherent risks of complications. A careful balance between preventive measures, early recognition, and appropriate interventions are critical to mitigate their impact. An adequate postoperative monitoring with rationale used of imaging studies, facilitates the timely identification of complications and guides for the optimal management.

CRediT authorship contribution statement

Camila Bras Harriott: Conceptualization, Methodology, Data curation, Writing – original draft, Visualization, Investigation. **Emmanuel E. Sadava:** Validation, Writing – review & editing, Supervision.

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